Hybrid Quantum Reinforcement Learning with Explainable AI for Dynamic Disaster Evacuation Route Optimization

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***Abstract*—Disater evacuation routes may become hazardous in a matter of minutes as flood levels, road availability, traffic, shelter availability, and quality of sensors change rapidly. A shortest path approach is able to adapt immediately but generally finds an optimal solution at one snapshot. Reinforcement learning can predict future state but even then an unregulated policy can suggest the use of a road which is closed or an overloaded shelter. In this paper, a hybrid quantum reinforcement learning model that distinguishes between adaptive route suggestion and safety enforcement. Deep reinforcement learning policy produces a few potential routes. A quadratic unconstrained binary optimization layer uses road closure, capacity, accessibility, priority, and resource constraints, where the Quantum Approximate Optimization Algorithm serves as an experimental solver, while a classical solver serves as a backup. Reliability of sources and contradictions between sources persist in the routing state. Explainable artificial intelligence layer integrates feature influence, route comparison, constraint information, and data provenance to produce an operator-friendly explanation. Stale or contradictory evidence triggers the system to halt automated dispatching and seek human intervention. Prototype experiments show that constraint filtering may significantly decrease unsafe selections at a marginal cost of clearance time. This paper makes no claim about quantum speedup; it provides an auditable architecture for adaptive, constraint-aware, and human-in-the-loop disaster evacuation routing.**

***Index Terms*—disaster evacuation, reinforcement learning, QAOA, QUBO, explainable artificial intelligence, dynamic routing, reliability-aware fusion, human-in-the-loop**

I. Introduction

Floods, wildfires, earthquakes, industrial incidents, and heavy storms may affect the usable road network during the course of evacuation [15]. What is initially a safe route can become obstructed by water, detritus, traffic congestion, or infrastructure damage. Evacuation routing is thus an ongoing process of decision making in the face of uncertainty.

An emergency system will rely on forecast data, river gauge readings, radar data, satellite imagery, road data services, mobile and other sensor readings, and reports from citizens. Not all of these data sources will fail for the same reasons. Gauges may experience drift, a radar image may become late, a road data service might provide old information, and an uncorroborated report may be either a timely warning or misinformation. To assume that all observations are of equal reliability could generate a path that is technically efficient but unsafe.

The classical routing algorithm is still useful because it is fast and transparent. Yet it responds to the current state of the graph and does not account for the sequential nature of the impact of congestion, shelters, and dangers. The RL algorithm learns about these sequential impacts, yet a learned policy does not ensure the feasibility of the solution. Quantum optimization tackles another aspect of the problem–route and shelters can be formulated as binary decisions with clear penalties. Present-day hardware does not make any general claims about the quantum speedup, so quantum optimization must be considered as just one solver algorithm in contrast to the only way out [6], [7].

These concepts are used together in a HQRL framework. RL suggests several good actions, and a safety QUBO layer picks up a feasible combination of them. The QAOA is applied to a limited number of solutions and the validated classical solver is run as a backup. This framework also considers the level of reliability of evidence, detects inconsistency of sources, explains why certain routes are chosen or discarded and requires an officer's approval for decisions taken in conditions

of high uncertainty. The four contributions are:

- A selection design that keeps adaptive learning distinct from hard constraints;
- A reliability-aware state where observations that are either outdated or contradictory impact routing and automation;
- A reasoning record that combines preference with concrete evidence of constraint violations; and
- An evaluation procedure involving route performance, dangerous routes, re-planning latency, failure recovery, and human validation.

not to replace the emergency officers, but to provide them with a quicker and more accountable decision aid.

II. Related Work

Flood susceptibility studies using machine learning employ tree ensembles, support vector machines, and artificial neural networks [1], [2]. The usefulness of forecasting on a larger scale and via satellite has been demonstrated in the operational services, yet the forecast is only a part of what determines which people need to evacuate, where they will go, and who will authorize the evacuation. Agent-based systems facilitate coordination of asynchronous processes performed by services and organizations [3].

The problem of evacuation routing was modeled as shortest path, quickest flow, minimum-cost flow, vehicle routing, and multiobjective optimization. The use of time-dependent graphs makes the model realistic in that it allows for varying travel time and capacity. The event of disaster adds a new kind of non-stationarity: an edge can become unavailable abruptly, and its availability is as important as its travel time.

Routing using RL is formulated as a Markov decision process (MDP) and is learned through interaction. Deep Q-networks allow for compact discrete decisions [4], and PPO can handle bigger action spaces and provide stable policy updates [5]. Routing and traffic control through RL has been adapted to varying demand, but ensuring safety becomes problematic. Reward penalties can help prevent violations in training while not guaranteeing that an illegal action will never happen in production.

QAOA alternates cost and mixing operations in its search for low energy binary solutions [6]. Assignment, capacity, and routing constraints can be represented in QUBO formulation [7]. Limitations include number of available qubits, sampling noise, optimizer sensitivity, and latency. This project relies on solver-agnostic QUBO layer and on exact or heuristic classical fall-back methods when possible.

Explainability of AI is also important for emergency scenarios. SHAP gives the additive contribution of each feature to the model output [8], but feature influence does not necessarily tell the officer why the path was rejected, e.g., due to closed bridge or lack of accessible shelter capacity. The proposed solution consists of multiple explanations: model attribution, contrastive routing data, constraint tracing, and source provenance.

Evidence fusion techniques form a good basis for representing support and uncertainty in heterogeneous observations [11]. Our technique brings novelty in terms of operational freshness, source fitness, and dispatch gating such that quality of evidence influences both route costs and authority to make decisions.

III. Problem Formulation

At the time of decision t, the impacted area is modeled as a directed graph $G_t = (V, E_t)$. Nodes include intersections, evacuation areas, shelters, and emergency resources. The dynamic vector of each edge e is denoted by

$$z_{e,t} = [\tau_{e,t}, d_e, q_{e,t}, h_{e,t}, c_{e,t}, \rho_{e,t}], \tag{1}$$

where τ is travel time, d is distance, q is congestion, h is hazard exposure, c is residual capacity, and ρ is information reliability. Closed edges are either eliminated or assigned a prohibitively high cost.

The system assigns groups $i \in I$ from their origin locations to compatible shelters $j \in J$. Each group has a size, priority, mobility requirement, and latest departure time. Shelters have finite capacity and might be limited in terms of need fulfillment capability. Availability of vehicles and limitations on roads form additional constraints.

The RL state includes the current state of the graph, hazard conditions, capacity limits of shelters and vehicles, reliability score, cross-source conflict, and previous actions. Action consists of selection of a route, shelter, dispatch priority, and a potential resource bundle. The reward is

$$r_t = -\alpha T_t - \beta H_t - \gamma Q_t - \delta U_t - \eta V_t + \zeta N_t, \tag{2}$$

where T_t is clearance time, H_t is exposure, Q_t is congestion, U_t is route churn, V_t is the number of violations, and N_t is the number of people evacuated. Churn is penalized since constant changes in the plan may disorient officers and evacuees.

The goal is not merely that of minimizing travel time. The system must move as many as possible of individuals to their safe capacity prior to hazard deadlines while ensuring protection of the vulnerable individuals. A recommendation becomes plausible if it does not violate closed edges, is within capacity limits of shelters, employs available resources, fulfills accessibility requirements, and complies with the approval policy that was configured.

IV. Reliability-Aware HQRL Algorithm

A. Observation Collection and Fusion

Source agents acquire meteorology, hydrology, radar, satellite, infrastructure, road, sensor, and crowdsourced observations. Each adapter adheres to a timeout, retry, cache, and fall-back contract. Schema, range, freshness, duplicate, and outlier checks are performed before the observation can affect routing state.

For each source s, its reliability becomes an inspectable weighted score

$$R_s = \omega_F F_s + \omega_V V_s + \omega_L L_s + \omega_A A_s + \omega_M M_s, \quad \sum_k \omega_k = 1. \tag{3}$$

where F_s indicates freshness, V_s validation success, L_s latency health, A_s historical agreement, and M_s operating mode. Observations made live get a better rating than cached or fallback observations. Reliability is not the same as confidence of a model: it is a measure of the quality of the evidence, not the confidence of a predictor.

Every source provides an opinion $y_{s,t}$ about a local hazard or road conditions on the road. Disagreement is measured by

$$C_t = \frac{\sum_{s<k} R_s R_k |y_{s,t} - y_{k,t}|}{\sum_{s<k} R_s R_k + \epsilon}. \tag{4}$$

Disagreement between two reliable sources is more important than an isolated unreliable observation. The graph contains both the estimate of the condition as well as its reliability.

B. RL Candidate Policy

PPO is selected when there is a shelter, route, priority, and number of dispatches in action; a deep Q-network could be used for smaller discrete set of candidates. Graph encoder is used to extract edge conditions, while context encoder is used for hazard, resource, and reliability variables. The training is done on a digital twin where rainfall, water level, road closure, congestion, demand, shelter loss, source delay, and communication failure [13] are varied.

Instead of giving only one irreversible action, the policy provides the top K candidates. This gives the long horizon advantage of the learning process but lets the safety layer reject the unsafe proposal. The rare closures and conflicts in sensors are deliberately kept during training, and the domain randomization is used by varying graph capacity, hazard evolution, compliance, and API delay.

The training procedure has three steps. In the first step, hazard and reliability features are configured. Then, the policy is pretrained under nominal traffic conditions and gradually introduced to sudden closure, correlated failure of sources, shelter unavailability, and noncompliance. The policy is finally plugged into the QUBO filter and trained on the reward after constraint satisfaction to avoid those candidates which are repeatedly filtered out.

C. QUBO Safety Layer

Let $x_{ik} = 1$ for group i choosing candidate k, and $y_{ij} = 1$ for being assigned to shelter j. A concise QUBO objective function is formulated by

$$\begin{aligned} H(x,y) = & \sum_{i,k} c_{ik} x_{ik} - \lambda \sum_{i,k} Q_\theta(s,i,k) x_{ik} \\ & + P_1 \sum_i \left(1 - \sum_k x_{ik}\right)^2 + P_2 \Phi_{\text{cap}}(y) \\ & + P_3 \sum_{i,k} b_{ik} x_{ik} + P_4 \Phi_{\text{res}}(x,y), \end{aligned} \tag{5}$$

where c_{ik} accounts for time, exposure, and congestion cost; Q_θ is the learned utility function; b_{ik} indicates blocked candidates; and Φ functions encode shelter capacity, accessibility, vehicle compatibility, and route overlap constraints. Slack variables are introduced to convert inequalities to binary quadratic functions. Penalty costs are set such that an infeasible gain will never beat a feasible one.

QAOA encodes the cost function using a cost Hamiltonian and searches for solutions among samples of bit strings; a toolkit such as Qiskit can enable the use of an experimental backend [14]. Each sample is decoded and validated deterministically. The QUBO is sent to a classical backend in case QAOA fails to satisfy the deadline or fails to provide a feasible assignment. The latter constitutes a safety mechanism, and not merely a means of improving performance.

Pruning of candidates is necessary. Graph loops, duplicates shelters, closed trips, as well as dominated solutions in both time and exposure are all eliminated prior to optimization. Geographic decomposition and rolling horizons reduce the size of the binary problem. It allows us to use the approach with classical methods today and to remain honest in small QAOA experiments.

D. Replaning, Explanation, and Human Approval

The system only replans when there is a real change, such as a new closure, hazard threshold crossing, loss of shelter capacity, increase in congestion, emergency vehicle request, or transition in reliability. The system uses a hysteresis period to prevent oscillation. Results are verified against the current version of the graph; in case of a graph change during the optimization process, the result is discarded and recalculated.

For any selected action, the explanation log includes information on the selected path, expected time and risk, reliability, conflicts, contributions from local features, bindings, provenance of source data, and the model and solver versions used. Based on this log, the interface can say why the long path was chosen due to the fact that the shorter one crosses a proven closure or why a certain group was rerouted due to the lack of capacity of a shelter. SHAP values are measures of influence rather than causation. The constraint traces are accurate only for the constraints that have been specified. Hence, explanations are provided as evidence for the decision rather than being a guarantee of the safety. In case the reliability drops below $R_{\min}$ or conflict rises above $C_{\max}$, the optimization will continue, but automatic dispatch will not be possible.

V. Experimental Design

The prototype is designed in a modular fashion with services for source acquisition, validation, reliability scoring, graph fusion, RL inference, QUBO formulation, solver, explanation, and audit logging. OpenStreetMap based and synthetically generated road networks are used to conduct route experiments. The public weather and flood data is used to test the hazard and feature pipelines. The digital twin is used to generate full episodes of evacuation which are not available in public tables [9], [10]. Population origins, vulnerable groups, shelter capacities, vehicles availability, compliance, congestion, hazard progression, road failures, source delay, and service outages make the episodes unique.

As per chaos engineering guidelines [12], the failure injection consists of weather API failure, delayed radar, sensor drift, out-of-date road network, citizen reports spam, database failure, and communication failure. Each failure propagates through reliability, conflict resolution, routing, explanation, and approval stages.

Methods for comparison include: static shortest path; time-dependent A*; greedy risk-aware heuristic; unconstrained PPO; PPO with a QUBO layer; and PPO with QAOA. The methods get the same graph updates. Metrics for the route include: clearance time, exposure, safe evacuation ratio, unsafe route ratio, capacity violations, replan delay, and route change. System metrics include: false dispatch; failure-recovery time; gating accuracy; and explanation delay.

Evaluation should separate training and test graphs as well as at least 30 seeded runs per scenario class. The disaster trajectories should be segmented by event rather than time step to avoid leakage. The gate is safety and capacity in a lexicographic manner: a method is considered deployable only after satisfying safety limits; the safe-evacuation rates and times are compared later. Ablations omit reliability scores, conflict gates, QUBO filters, and route explanations to determine which element is responsible for each difference.

A. Runtime Algorithm and Safety Conditions

In runtime, source agents perform updates asynchronously; however, routing decisions are always made based on an immutable snapshot of the state. The snapshot includes a time stamp, graph version, source modes, reliability scores, and provenance id. Policy computes its candidate set based on an inference budget. Candidate pruning removes cycles, duplicate shelters, closed edges, and suboptimal candidates by both time and exposure. The trimmed-down problem is then sent concurrently to the preferred QUBO backend and a classical fallback. A solution is never sent unverified. A deterministic checker verifies that the selected routes fit the current graph, shelter capacities, vehicle stocks, and accessibility policies.

If the graph changes during the optimization process, the result becomes stale and is discarded. The optimistic concurrency principle guarantees that a route calculated prior to closing of the edge will not be suggested as an option on the approval screen. This approach increases reproducibility since each recommendation can be uniquely associated with one versioned input state.

TABLE I
Dynamic Routing Results Representative Pattern

Method	Clear. (min)	Safe (%)	Unsafe (%)	Replan (ms)
Static shortest path	47.8	82.1	11.7	18
Time-dependent A*	41.6	88.4	6.8	46
Risk-aware greedy	40.9	90.2	4.9	31
PPO	37.2	92.5	3.8	22
PPO + classical QUBO	38.0	96.4	0.6	74
PPO + QAOA	38.3	96.1	0.7	486

TABLE II
Failure Injection Behavior Representative Pattern

Condition	ΔR	Ungated unsafe (%)	HQRL unsafe (%)
Weather API outage	-0.12	5.4	1.1
Delayed radar	-0.09	4.8	0.9
Gauge drift	-0.17	8.6	1.5

Latency budget is split between acquisition, feature construction, RL inference, optimization, explanation, and interface delivery. QAOA is not allowed to use the entire budget for emergency responses. Deadline for QAOA is chosen based on the urgency of the event, and timeout will promote the validated classical solution. Audit log indicates if the suggested action was generated by QAOA, classical QUBO solver, greedy solution, or if it was a previously safe plan. It matters where the solution comes from since two solutions that appear the same can have different validation stories.

Episodes end when all groups are safe, the hazard deadline passes, or there is no path left. Rare closures and correlated failures are over-represented during training while evaluation keeps their natural ratio. A minimum service duration is set to prevent the policy from optimizing mean clearance time by constantly postponing the same small vulnerable group. Because of that, clearance time of the worst served zone is also reported.

VI. Prototype Results and Discussion

Table I and Table II provide example values from the seeded prototype which exhibit the evaluation pattern. They are not results from a field experiment and must be re-run from an archived implementation before submission. The example pattern exhibits the expected trade-off. PPO decreases clearance time by planning for congestion, but an unbounded policy sometimes picks routes which fail a future closure or capacity test. QUBO filtering reduces unsafe choices significantly at the cost of only a modest decrease in clearance time. The quality of the solutions produced by QAOA and the classical solver are equivalent on small problem instances, while the simulation of the QAOA route is slower.

These numbers justify the use of hybrid constraint enforcement, not any advantage in quantum speed. The failure injection test reveals a limitation of normal benchmarks for routing algorithms. When the information about the road

network goes out of date, quick computation does not mean good choice. Reliability propagation decreases confidence in edge reliability, conflict detection detects conflicts with other sources of information, and the human gate stops dispatch when doubt exceeds certain threshold. The process takes more time to approve the dispatch, but the reason is clear.

Take, for example, a situation where there are floods affecting three low areas and two shelters. First, the learned policy prefers a short arterial route. However, the QUBO layer determines that all the groups to be transported to one shelter would not fit into the transport's capacity; hence, the mobility dependent group goes to the second shelter. Afterwards, a water sensor determines that there is an extremely fast depth development in the arterial despite the road service still calling it open. There is conflict here, but it does not close the edge at first since the sensor lacks a long history. It closes the edge when both the radar and geolocated reports confirm the findings of the sensor, and the safety layer deletes the original RL recommendation. Only the groups who haven't passed the last safe diversion point are rerouted.

Scalability is the key computational issue. In the presence of I evacuation groups, K saved routes, and J shelters, the safety level has roughly $IK + IJ$ binary variables initially. Citywide representation is unfeasible for present-day quantum computers and even for exact classical algorithms. An application in the real world would require the splitting of the area into zones, coordination of arterial roads and shelters, and constrained optimization only if groups compete for limited resources.

A. Ablation and Sensitivity Analysis

Component-level evaluations are required because the unsafe-route ratio can stem from the learned policy, the constraint filter, or the approval gate. Reliability ablation provides equal trust to all sources. Conflict ablation ensures that a discrepancy cannot impact dispatch responsibility. The safety-layer ablation follows the pure RL action, whereas the explanation ablation suppresses the route rationale without changing the decision. The contrast of those configurations allows to differentiate between improvements in prediction and governance or feasibility enforcement.

Reliability and conflict thresholds have to be swept together as they govern a practical trade-off, as high thresholds decrease automatic dispatching while increasing the officer's workload. Reward coefficients and QUBO penalties also have to be varied within wide ranges. The penalty should remove infeasible solutions without obscuring the differences between feasible routes. QAOA sensitivity analysis involves circuit depth, number of shots, optimizer, initialization and noise. RL sensitivity analysis includes number of candidates, graph encoder size, discount factor, entropy regularization and hysteresis in replanning. Distribution results per seed are more valuable than picking the best run out of multiple.

The results also have to be understood in an operational way. If a small increase in the clearance time helps in overcoming the problem of capacity violation, it is fine; however, this same delay can turn out to be detrimental in case of a rapidly approaching danger. Thus, metrics should be presented on the basis of the severity of the event and the demographic of the people involved and not as an average figure. The approval time must be segregated from the algorithmic latency time.

VII. Deployment, Constraints, and Ethics

Deployment must follow three levels of decision-making. Standard and low-risk movements can utilize deterministic routing. RL solutions are triggered only if altering circumstances of congestion and hazards make sequential adaptation worthwhile. Constrained optimization is employed in scenarios where there is competition between the groups for road, shelter, or dedicated vehicles. The order of fallbacks is a validated hybrid solution, a classical constrained solution, a greedy safe solution, a previously validated solution, and a human-controlled one.

At the outset, the system must be run in shadow mode with live observations and recommendations made but not sent to the field units. The advisory mode may start to be used once certain criteria regarding safety, calibration, latency, and usability are fulfilled. Any automatic control must be limited to low-risk decisions and reversible via a policy setting.

In a given situation, the decision card given to the officer should be succinct compared to an extensive dashboard of technical information. The card will have all the information on the affected parties, chosen route, best alternative, reason for refusal, delay, capacity of the shelter, age of the evidence and need for approval. The officer can either accept, change or reject the decision. The process will be documented with a clear explanation of what the officer saw, hence differentiating a model error from human input.

This work is an architecture and prototype contribution. It does not demonstrate quantum advantage, develop a new RL convergence theorem, or replace hydrodynamic flood modeling. Synthetic scenarios cannot completely capture informal roads, human non-compliance, communications disruption, or institutional limitations. Reliability coefficients and permission thresholds are expert-set initially and risk undue faith in a particular authority or silencing a valuable dissenting view. They need to be tuned based on actual results.

Optimization risks putting at a disadvantage communities which are distant, have low mobility, or lack connectivity if mean travel time is the only metric used. Prioritization criteria must make explicit provisions for hospitals, seniors, kids, the disabled, and households without cars. Civic reporting needs to include privacy protections, authentication, rate throttling, and bias prevention. Audit logs need to be tamper-proof, access-controlled, and retained as per governance policy.

No pathway can be said to have guaranteed safety. The visualization needs to indicate the timestamp on evidence, whether a source is current or cached, what assumptions are being made in the plan, and what options are left. Ethics cannot be ensured by approval alone; the officers will also require training, authority, alarms that are feasible, and incident analysis afterwards.

Future research needs to employ online Bayesian calibration and conformal prediction for coverage-aware uncertainty instead of fixed reliability weights. The evaluation should cover flood replay from archives, calibrated hydrodynamic modeling, emergency operations drills, and usability tests. The quantum experiments should mention circuit depth, qubit number, shot count, approximation ratio, optimizer configuration, noise model, and strong classical baselines. Verification of the dispatch gating state machine would be an additional step to increase safety guarantees.

VIII. Conclusion

This work has developed a reliability-based HQRL system for adaptive disaster evacuation routing, with RL providing flexible route proposals, and a solver-agnostic QUBO layer ensuring closures, capacity, accessibility, prioritization, and resource constraints. QAOA remains used as a backend experiment with a classical algorithm for fall back, to prevent any claim of quantum superiority that is not proven yet. Explanation layer combines the influence of features with route comparison, exact constraint violation, source and proof of the evidence, as well as human approval, thus making the approach to the problem as a chain of trust-governed decisions rather than shortest path problem. Further field verification, reliability evaluation, and institutional governance will be necessary for the use in practice.

Acknowledgment

The authors would like to thank providers of the Open-Meteo, NASA POWER, NASA GIBS, Copernicus Sentinel, OpenStreetMap and OSRM data.

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